



Chapter	Title	Total questions	Type	Date
Ch 05	Prime Time	11	Formative	23 May 2026

SECTION A — RECALL AND APPLY

Q1. In the Idli-Vada game, players say 'idli' for multiples of 3 and 'vada' for multiples of 5. At which number do players say 'idli-vada' for the very first time?

Section: section 5.1 · Type: MCQ · Goal: Recall

- A. 8
- B. 15
- C. 18
- D. 30

Exercise — Figure it Out Q1, section 5.1 p.108

At what number is 'idli-vada' said for the 10th time? (Find the 10th common multiple of 3 and 5.)

Q2. A prime number is defined as a number that has exactly two factors: 1 and itself. Which of the following statements about the number 1 is correct according to this definition?

Section: section 5.2 · Type: MCQ · Goal: Recall

- A. 1 is a prime number because it is not divisible by any number other than itself.
- B. 1 is a composite number because it can be arranged in a rectangle.
- C. 1 is neither prime nor composite because it has only one factor.
- D. 1 is prime because it is the smallest natural number.

Exercise — Figure it Out Q1, section 5.2 p.114

Is there any even prime other than 2?

Q3. Two numbers are called co-prime if their only common factor is 1. Which of the following pairs is co-prime?

Section: section 5.3 · Type: MCQ · Goal: Recall

- A. 12 and 18
- B. 9 and 15
- C. 8 and 21
- D. 14 and 35

Exercise — Figure it Out Q1, section 5.3 p.115

Check if each pair is 'safe' (co-prime): (a) 15 and 39; (b) 4 and 15; (c) 18 and 29; (d) 20 and 55.

SECTION B — REASON AND EXPLAIN

Q4. In the Sieve of Eratosthenes, after circling 2 and crossing out all its multiples, the next step is to circle 3 and cross out all its multiples. A classmate says: 'We can stop after crossing out multiples of 5, because any number greater than 5 that is not yet crossed out must be prime.' Do you agree or disagree? Justify your answer using what you know about prime numbers and factors.



Section: section 5.2 · Type: ECR · Goal: Reason

Exercise — Figure it Out Q3, section 5.2 p.114

Are there an equal number of primes in every row of the sieve table? Which decades have the fewest primes? Which have the most?

Q5. Anshu writes $84 = 4 \times 21$ and concludes that 84 and 55 are co-prime because neither 4 nor 21 appears as a factor of 55. Explain why Anshu's reasoning is wrong and show the correct way to check co-primality using prime factorisation.

Section: section 5.4 · Type: SCR · Goal: Reason

Exercise — Figure it Out Q1, section 5.4 p.122

Are the following pairs co-prime? Guess first, then verify using prime factorisation: (a) 30 and 45; (b) 57 and 85; (c) 121 and 1331; (d) 343 and 216.

Q6. Meena says: 'If a number is divisible by 8, it must also be divisible by 4 and by 2.' Priya says: 'If a number is divisible by 2 and by 4, it must be divisible by 8.' Decide which statement is always true and which is sometimes false, and justify your answer using the divisibility tests from this chapter.

Section: section 5.5 · Type: SCR · Goal: Reason

Exercise — Figure it Out Q3, section 5.5 p.125

Explore whether each statement is always, sometimes, or never true: (a) sum of two even numbers is a multiple of 4; (b) sum of two odd numbers is a multiple of 4.

SECTION C — APPLY IN CONTEXT

Q7. In a Jump Jackpot game, Grumpy has hidden treasures at the numbers 36 and 48. Jumpy wants to choose a single jump size greater than 1 so that, starting from 0, she eventually lands on both treasure numbers. List all the jump sizes that work, and identify the largest possible jump size.

Section: section 5.1 · Type: NUM · Goal: Apply

Exercise — Figure it Out Q1, section 5.1 p.111

In Jump Jackpot, treasures are on 28 and 70. What jump sizes land on both numbers?

Q8. In the Co-prime Art activity, a thread is wound around pegs placed equally on a circle. Starting at peg 0, the thread jumps 4 pegs at a time on a circle with 9 pegs numbered 0 to 8. (a) Will every peg be visited before the thread returns to peg 0? (b) Now try a jump of 3 pegs on the same 9-peg circle. Will every peg be visited? Explain both answers using the idea of co-prime numbers.

Section: section 5.3 · Type: SCR · Goal: Apply

Exercise — Activity (Co-prime Art), section 5.3 p.116

Students observe and analyse thread-art diagrams with different peg counts and thread-gaps....



Q9. Use prime factorisation to answer both parts. (a) Find the prime factorisation of 180. (b) Is 180 divisible by 45? Show your working using the prime factorisations of both numbers.

Section: section 5.4 · Type: NUM · Goal: Apply

Exercise — Figure it Out Q1, section 5.4 p.120

Find the prime factorisations of: 64, 104, 105, 243, 320, 141, 1728, 729, 1024, 1331, 1000.

Q10. A school plans a sports day and wants to divide 5,840 students into equal groups. Without performing long division, determine whether 5,840 is divisible by each of the following: (a) 2, (b) 5, (c) 4, (d) 8. Apply the divisibility tests from this chapter and show the check for each.

Section: section 5.5 · Type: NUM · Goal: Apply

Exercise — Figure it Out Q1, section 5.5 p.125

2024 is a leap year. Leap years are multiples of 4, except multiples of 100 that are not multiples of 400....

Q11. Look at the following group of four numbers: 11, 15, 17, 23. Each number can be considered 'special' compared to the other three for a different reason. For any two of the four numbers, write what property makes that number stand out from the rest. (There may be more than one valid reason - any well-explained answer using ideas from this chapter is acceptable.)

Section: section 5.6 · Type: SCR · Goal: Apply

Exercise — Math Talk (Special Numbers), section 5.6 p.126

In each box of four numbers, explain how each number is special compared to the others....

Notes

1. Marks are not prescribed — assign per question based on formative assessment weightage and cognitive demand.
2. Teacher guidance (expected answer, method, distractors, inclusivity) is available in the online view under My Plans > Chapter Assessment.